

SESSION 4

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1. Large Group Positions

CODING

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

/**
 * Return an array of arrays of size *returnSize.
 * The sizes of the arrays are returned as *returnColumnSizes array.
 */
int** largeGroupPositions(char* s, int* returnSize, int** returnColumnSizes) {
    int n = strlen(s);
    int capacity = 10;
    int** result = (int**)malloc(sizeof(int*) * capacity);
    *returnColumnSizes = (int*)malloc(sizeof(int) * capacity);

    int count = 0;
    int start = 0;

    for (int i = 0; i <= n; i++) {
        if (i == n || s[i] != s[start]) {
            if (i - start >= 3) {
                if (count >= capacity) {
                    capacity *= 2;
                }
                result[count] = (int*)malloc(sizeof(int) * (i - start));
                (*returnColumnSizes)[count] = i - start;
                for (int j = start; j < i; j++) {
                    result[count][j - start] = s[j];
                }
                count++;
            }
            start = i;
        }
    }
    *returnSize = count;
    return result;
}
```

```

        result = (int**)realloc(result, sizeof(int*) * capacity);
        *returnColumnSizes = (int*)realloc(*returnColumnSizes, sizeof(int) * capacity);
    }
    result[count] = (int*)malloc(sizeof(int) * 2);
    result[count][0] = start;
    result[count][1] = i - 1;
    (*returnColumnSizes)[count] = 2;
    count++;
}
start = i;
}
}

*returnSize = count;
return result;
}

int main() {
    char s[] = "abbxxxzzy";
    int returnSize;
    int* returnColumnSizes;

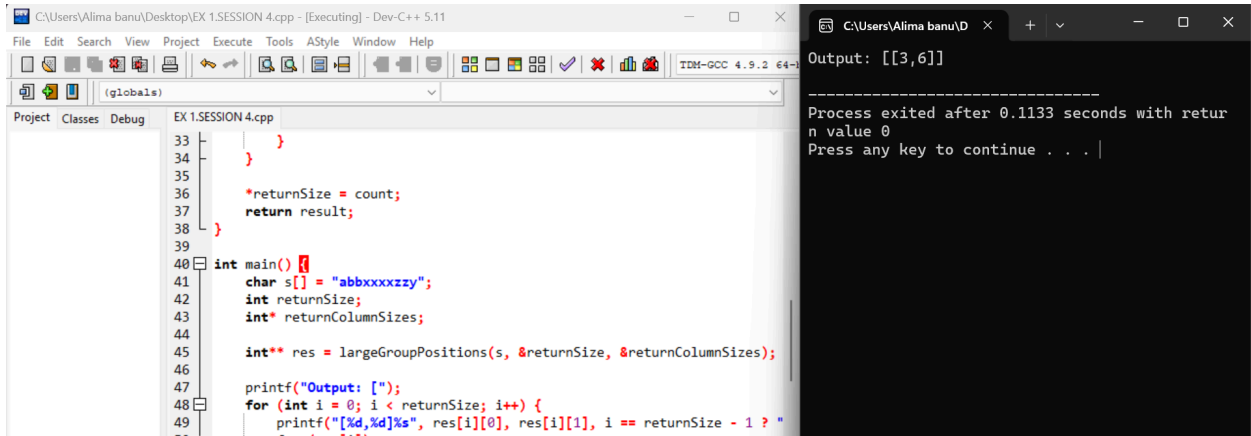
    int** res = largeGroupPositions(s, &returnSize, &returnColumnSizes);

    printf("Output: [");
    for (int i = 0; i < returnSize; i++) {
        printf("[%d,%d]%", res[i][0], res[i][1], i == returnSize - 1 ? "" : ",");
        free(res[i]);
    }
    printf("]\n");

    free(res);
    free(returnColumnSizes);
    return 0;
}

```

OUTPUT:



The screenshot shows a Dev-C++ IDE window titled "C:\Users\Alima banu\Desktop\EX 1.SESSION 4.cpp - [Executing] - Dev-C++ 5.11". The code editor displays a C++ program with the following visible lines:

```
33     }  
34 }  
35  
36 *returnSize = count;  
37 return result;  
38 }  
39  
40 int main()  
41 {  
42     char s[] = "abbxxxxzzy";  
43     int returnSize;  
44     int* returnColumnSizes;  
45  
46     int** res = largeGroupPositions(s, &returnSize, &returnColumnSizes);  
47  
48     printf("Output: [");  
49     for (int i = 0; i < returnSize; i++) {  
50         printf("[%d,%d]", res[i][0], res[i][1]);  
51         if (i == returnSize - 1) printf("]");  
52     }  
53 }
```

The output window on the right shows the following text:

```
Output: [[3,6]]  
  
-----  
Process exited after 0.1133 seconds with return value 0  
Press any key to continue . . .
```

2.Game of Life

```
#include <stdio.h>  
#include <stdlib.h>
```

```
/*
```

Bit Manipulation Trick for In-Place Update:

- 00 (0): dead -> dead
- 01 (1): live -> dead
- 10 (2): dead -> live
- 11 (3): live -> live

Bit 0 represents the CURRENT state.

Bit 1 represents the NEXT state.

```
*/
```

```
void gameOfLife(int** board, int boardSize, int* boardColSize) {  
    int m = boardSize;  
    int n = boardColSize[0];
```

```
    int directions[8][2] = {  
        {-1, -1}, {-1, 0}, {-1, 1},  
        { 0, -1},      { 0, 1},  
        { 1, -1}, { 1, 0}, { 1, 1}  
    };
```

```

for (int r = 0; r < m; r++) {
    for (int c = 0; c < n; c++) {
        int live_neighbors = 0;

        for (int d = 0; d < 8; d++) {
            int nr = r + directions[d][0];
            int nc = c + directions[d][1];

            if (nr >= 0 && nr < m && nc >= 0 && nc < n) {
                live_neighbors += (board[nr][nc] & 1); // Get current state
            }
        }

        // Rules application
        if ((board[r][c] & 1) == 1) {
            if (live_neighbors == 2 || live_neighbors == 3) {
                board[r][c] |= 2; // Next state becomes 1 (binary 11 = 3)
            }
        } else {
            if (live_neighbors == 3) {
                board[r][c] |= 2; // Next state becomes 1 (binary 10 = 2)
            }
        }
    }
}

// Update board to next state
for (int r = 0; r < m; r++) {
    for (int c = 0; c < n; c++) {
        board[r][c] >>= 1;
    }
}

int main() {
    int m = 4, n = 3;
    int grid[4][3] = {
        {0, 1, 0},
        {0, 0, 1},
        {1, 1, 1},
        {0, 0, 0}
    };
};

```

```

int* board[4];
int colSize[4] = {3, 3, 3, 3};
for (int i = 0; i < 4; i++) board[i] = grid[i];

gameOfLife(board, m, colSize);

printf("Output:\n");
for (int r = 0; r < m; r++) {
    for (int c = 0; c < n; c++) {
        printf("%d ", board[r][c]);
    }
    printf("\n");
}
return 0;
}

```

OUTPUT:

```

C:\Users\Alima banu\Desktop\EX 2.S54.cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
(globals)
Project Classes Debug EX 1.SESSION 4.cpp EX 2.S54.cpp
66      };
67
68      int* board[4];
69      int colSize[4] = {3, 3, 3, 3};
70      for (int i = 0; i < 4; i++) board[i] = grid[i];
71
72      gameOfLife(board, m, colSize);
73
74      printf("Output:\n");
75      for (int r = 0; r < m; r++) {
76          for (int c = 0; c < n; c++) {
77              printf("%d ", board[r][c]);
78          }
79          printf("\n");
80      }
81      return 0;
82
Output:
0 0 0
1 0 1
0 1 1
0 1 0

-----
Process exited after 0.1149 seconds with return value 0
Press any key to continue . . .

```

3.Champagne Tower

CODING:

```
#include <stdio.h>
```

```

double champagneTower(int poured, int query_row, int query_glass) {
    double dp[101][101] = {0.0};
    dp[0][0] = (double)poured;

```

```

for (int r = 0; r <= query_row; r++) {
    for (int c = 0; c <= r; c++) {
        if (dp[r][c] > 1.0) {
            double overflow = (dp[r][c] - 1.0) / 2.0;
            dp[r][c] = 1.0;
            if (r + 1 <= query_row) {
                dp[r + 1][c] += overflow;
                dp[r + 1][c + 1] += overflow;
            }
        }
    }
}

return dp[query_row][query_glass] > 1.0 ? 1.0 : dp[query_row][query_glass];
}

int main() {
    printf("Output: %.5f\n", champagneTower(1, 1, 1)); // Output: 0.00000
    printf("Output: %.5f\n", champagneTower(2, 1, 1)); // Output: 0.50000
    return 0;
}

```

OUTPUT:

The screenshot shows the Dev-C++ IDE with the following code in EX 3.SS4.cpp:

```

11         dp[r][c] = 1.0;
12         if (r + 1 <= query_row) {
13             dp[r + 1][c] += overflow;
14             dp[r + 1][c + 1] += overflow;
15         }
16     }
17 }
18
19
20 return dp[query_row][query_glass] > 1.0 ? 1.0 : dp[query_row][query_glass];
21
22
23 int main() {
24     printf("Output: %.5f\n", champagneTower(1, 1, 1)); // Output: 0.00000
25     printf("Output: %.5f\n", champagneTower(2, 1, 1)); // Output: 0.50000
26     return 0;

```

The output window shows the following text:

```

Output: 0.00000
Output: 0.50000

-----
Process exited after 0.1222 seconds with return value 0
Press any key to continue . . .

```